AUTOMATION PORTFOLIO — TECHNICAL RECORD

SOPHON FINANCE SYSTEMS

AI-Driven Finance & Accounting Automation | github.com/sophonfinance-wq/finance-automation-portfolio

PARTNERSHIP 1065 / SECTION 704(C)

A deterministic, AI-assisted Form 1065 partnership-tax engine that builds a book-to-tax bridge, maps return lines, allocates K-1s to the exact penny, and runs tie-out checks — with an add-on IRC §704(c) built-in-gain module — on fully fictional, seeded data.

Report No.	SFS-ENG-004	**Package**	partnership_tax
System	Partnership 1065 / Section 704(c)	**Data**	Fictional / seeded
Run	`python -m partnership_tax`	**Status**	Public — portfolio record

1.0 OVERVIEW AND SCENARIO

A fictional real-estate development partnership — the seeded Demo 721 Development LP — must file Form 1065 with a corporate general partner holding 0.01% and two LLC limited partners at 89.99% and 10.00%. The engine bridges $615,550.00 of book income to $653,550.00 of ordinary business income (adding back nondeductible syndication costs and book depreciation, then deducting tax depreciation) and splits that income across the three partners without the K-1 amounts drifting from the Schedule K total — the largest-remainder allocator absorbs the odd penny so the K-1-to-Schedule-K check (CHK-002) holds at a 0.00 difference. In the companion §704(c) demo (Harborview Partners LP), where Atlas Capital LLC contributes a building with a $900,000.00 built-in gain against Beacon Equity LLC's cash, the engine detects and flags the traditional-method ceiling rule binding $50,000.00 each year rather than silently curing the distortion, then catches up the residual $150,000.00 layer to the contributor on sale.

2.0 WHAT THE SYSTEM AUTOMATES

- Source intake from a seeded fictional bundle: trial balance / P&L, balance sheet, member capital accounts, syndication-cost support, and a prior-return line map
- Book-to-tax bridge that reconstructs book income and applies Schedule M-1 adjustments (nondeductible syndication costs, book-depreciation addback, tax-depreciation deduction) to derive ordinary business income
- Return-line mapping to Form 1065 (lines 1c, 14, 20, 21, 22), Schedule K, Schedule L, Schedule M-1, and Schedule M-2, with a source ID attached to every mapped line
- Partner K-1 allocation of ordinary income plus a capital-account rollforward (BOY, contributions, distributions, EOY) using exact integer-cent, largest-remainder allocation so K-1 amounts sum precisely to Schedule K
- A deterministic reviewer tie-out check registry (CHK-001 through CHK-007) covering return-line agreement, K-1 sums, Schedule L balance, M-1/M-2 reconciliation, partner-percentage total, and source-reference completeness
- Review-package emission in Markdown (tax_workpapers.md, review_checks.md) and JSON (form_1065_preview.json), plus an optional locally generated Excel support workbook

- An optional IRC §704(c) built-in-gain module: parallel §704(b) book and tax capital accounts, traditional-method tax-depreciation allocation, ceiling-rule detection, on-sale catch-up, and per-partner K-1 capital analysis on both bases

3.0 OPERATING PROCEDURE

1. Generate or ingest the seeded fictional source package (partners, income items, deduction items, book-tax adjustments, balance sheet, capital activity)
2. Compute book income as total income minus book deductions, then add book-tax adjustments to derive ordinary business income (all in integer cents)
3. Build the Form 1065 / Schedule K / Schedule L / Schedule M-1 / Schedule M-2 line map, tagging each line with its supporting source IDs
4. Allocate ordinary income across partners by profit basis points using largest-remainder integer-cent allocation, and roll each partner's capital account forward to ending capital
5. Run the review-check registry, comparing expected vs. actual for each of the seven checks at zero tolerance
6. Write tax_workpapers.md, review_checks.md, and form_1065_preview.json (plus optional xlsx), marking the package READY only if every check passes
7. For the §704(c) demo, seed formation capital at FMV (book) and carryover basis (tax), then roll each fiscal year applying traditional-method depreciation with ceiling-rule flags
8. On sale, catch up the residual §704(c) layer to the contributing partner and emit the §704(c) summary plus one per-partner K-1 Markdown file
9. Return CLI exit code 0 when the package is READY, or 2 when review is needed

4.0 CONTROLS AND SAFEGUARDS

- Fully fictional, seeded data only, with a disclaimer embedded in every output ('No real taxpayer, partner, EIN, or client figure is included')
- Exact integer-cent money model with largest-remainder allocation, so partner K-1 allocations tie exactly to the Schedule K total even when percentages create pennies
- Source ID attached to every mapped return line, with CHK-007 verifying that all mapped lines carry a source reference
- Deterministic zero-tolerance tie-out identities: Form 1065 line 22 = Schedule K line 1; K-1 sum = Schedule K; Schedule L balances; M-1 and M-2 reconcile; partner percentages sum to 100%
- In the §704(c) module the ceiling-rule distortion is surfaced (flagged BINDING with the uncured shortfall) rather than silently cured, and the traditional-method-only limitation is documented by design
- Human review remains the final sign-off; the package is auto-marked READY only when all checks pass, otherwise the CLI signals REVIEW via exit code 2

5.0 VERIFICATION AND TESTING

Correctness is proven by seven zero-tolerance tie-out checks (CHK-001 through CHK-007), whose expected/actual differences are all 0.00 in the sample review_checks.md, and by the §704(c) tax-basis balance sheet that ties out (assets $2,260,000.00 = liabilities $0 + tax capital $2,260,000.00). The engine ships with a curated pytest suite the README states as 1,605 tests, implemented across 185 test functions in ten modules including parametric grids, engine-pipeline coverage, and a dedicated §704(c) suite. The CLI returns exit code 0 only when every review check passes and exit code 2 otherwise, so a broken tie-out fails the run rather than silently producing a package.

6.0 KEY FACTS

Attribute	Value
Curated tests (README)	1,605
Test functions / modules	185 across 10 modules
Reviewer tie-out checks	7 (CHK-001–CHK-007)
Standard-demo ordinary income (fictional)	$653,550.00
§704(c) built-in gain at formation (fictional)	$900,000.00
Partners in 1065 demo	3 (1 GP + 2 LP)

This record documents a self-contained system in the Sophon Finance Systems automation portfolio. All data is fictional and seeded; no employer or client workpaper, entity, methodology, path, or figure is reproduced. Prepared 14 July 2026.